



Vivek Mishra <vivekmishraima@gmail.com>

GSP2025 : Paper 17

1 message

Microsoft CMT <email@msr-cmt.org>

Sat, Mar 22, 2025 at 12:30 PM

Reply-To: Microsoft CMT - Do Not Reply <noreply@msr-cmt.org>

To: Vivek Mishra <vivekmishraima@gmail.com>

Vivek Mishra has uploaded review for Differentially Private Decentralized PCA of Graph Signals for Graph Clustering

-- Review Summary --

1. Summary

The paper introduces a decentralized PCA method for graph clustering while ensuring differential privacy. It addresses the challenge of agents having access to only partial graph signals and proposes the Decentralized and Differentially Private Power Method (Dec-DP-Pow) to compute principal eigenvectors securely. By leveraging consensus aggregation, the approach minimizes communication overhead while preserving privacy. Theoretical (ϵ, δ) -differential privacy guarantees are established, and its effectiveness is demonstrated through numerical experiments.

2. Strengths

The proposed method minimizes communication overhead by allowing each agent to track only its local portion of the eigenvectors, reducing data sharing. Its effectiveness is validated through numerical experiments on a stochastic block model graph. Theoretical guarantees are established using Chernoff bounds and composition theorems for differential privacy. Additionally, the privacy-preserving mechanism ensures sensitive graph signals remain protected while enabling accurate clustering.

3. Weaknesses

The paper's scalability remains uncertain as it is tested on a small-scale setup (400 nodes, 4 agents), leaving its performance on larger graphs unexplored. The added differential privacy mechanism may introduce computational overhead, but its impact on efficiency is not thoroughly analyzed. Experimental evaluation is limited, primarily comparing decentralized clustering with centralized k-means, lacking comparisons with other DP-PCA methods. Additionally, the approach assumes graph signal smoothness, which may not be applicable to all real-world datasets.

4. Questions

Can you apply this method to real world dataset like social media?

5. Rating

Accept

To stop receiving conference emails, you can check the 'Do not send me conference email' box from your User Profile.

Microsoft respects your privacy. To learn more, please read our [Privacy Statement](#).

Microsoft Corporation
One [Microsoft Way](#)
[Redmond, WA 98052](#)